



Fig. 13: The resultant image using Sobel operator for edge detection



Fig. 14: Image with Gaussian noise processed using the Sobel edge detector



Fig. 15: Noisy image

pixel values using the convolution operator. Instead, these usually apply a ranking (sorting) function to pixel values within the neighbourhood and select a value from the sorted list. For this reason, these are sometimes called rank filters. Examples of non-linear filters include median filter and max and min filters.

Median filter is a popular non-linear filter used in image process-



Fig. 16: Cleaned image



Fig. 17: Another noisy image



Fig. 18: The image cleaned using median filter

ing. It works by sorting the pixel values within a neighbourhood, finding the median value, and replacing the original pixel value with the median of that neighbourhood. Median filter works much better than an averaging filter with comparable neighbourhood size in reducing salt and pepper noise. Because of its popularity, median filter has its own function (`medfilt2`) provided by the IPT.

Following code shows the use of median filter to remove salt and pepper noise from the image `Penguins_grey.jpg`:

```
A = imread('Penguins_grey.jpg');
Noise_snp=imnoise(A,'salt&pepper');
imshow(Noise_snp)
Filtered_snp=medfilt2(Noise_snp,[3 3]);
imshow(Filtered_snp)
```

Figs 15 and 16 show noisy and cleaned images, respectively. Fig. 17 shows another noisy image and Fig. 18 shows this image cleaned using median filter.

## Sliding window operations

Sliding window neighbourhood operations are implemented in the IPT using one of these two functions: `nlfilter` or `colfilt`. Both functions accept a user-defined function as a parameter. Such function can perform linear as well as non-linear operations on the pixels within a window. For rank filters, the IPT function `ordfilt2` makes it very easy to create min, max and median filters.

The syntax for `nlfilter` has three arguments: the image matrix, the size of the filter and the function to be applied. The following line defines a filter that takes the maximum value in a 3x3 neighbourhood:

```
cmax = nlfilter(c,[3,3],'max(x(:))');
```

A filter that takes minimum value in a 3x3 neighbourhood is defined by the following command:

```
cmin = nlfilter(c,[3,3],'min(x(:))');
```

## Region of interest processing

Filtering operations are sometimes performed only in a small part of an image, referred to as the region of interest (ROI). ROI is specified by defining a mask that limits the portion of the image in which the operation will take place.

Image masking is the process of extracting a sub-image from a larger image for further processing. ROI processing can be implemented in MATLAB using a combination of two functions: `roipoly` (for image masking) and `roifilt2` (for actual processing of the selected ROI).

To be continued next month